<table>
<tr><td>Module Title:</td><td>Foundations of Cyber Security</td></tr>
<tr><td>Level:</td><td>MSc</td></tr>
<tr><td>Assessment title:</td><td>Preserving file integrity (individual work: 60%)</td></tr>
<tr><td>Deadline:</td><td>Friday 17/10/2025 13:00</td></tr>
<tr><td>Assessed intended learning outcomes:</td><td>
Demonstrate ability to produce secure code to a specification.
Demonstrate a deep understanding of fundamental cryptographic concepts such as hash functions and applying them to preserve and verify file integrity.
Demonstrate ability to apply the OpenSSL cryptography library to solve security goals.
Learn to handle errors gracefully and securely, especially in cryptographic operations where errors can have severe consequences.
Gain practical experience in using the cryptographic library OpenSSL.
Learn to communicate complex technical topics.
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<tr><td>Weighting within module:</td><td>This assessment is worth 60% of the overall module mark.</td></tr>
<tr><td>Task details and instructions:</td><td>This assessment task requires you to implement a command-line tool in C that reads a file, computes its cryptographic hash, and outputs the results in hexadecimal format.

The tool should accept a user-provided hash and verify whether the computed hash matches the expected value, indicating whether the file's contents have been modified.

Your tool should be simple to use and support at least two hashing algorithms. One algorithm should be set as the default, while the other algorithm(s) can be selected by the user through a command-line.

The tool should operate as <<filehash -a <algorithm> <filename>>. The hashing algorithm options should be displayed so that users can choose their desired algorithm, if none is chosen, then the default one is used.

If only a filename is provided, the program outputs the computed hash. If an expected hash is also provided, the program should compare it to the computed value and report whether the integrity check passes or fails.

Your program should adhere to the following specifications:
It must be implemented in C and utilize the OpenSSL cryptography library.
The program should display available algorithm options to users.
At least two cryptographic hashing algorithms must be implemented.
If no algorithm is specified, the program must use the default algorithm.
If only a filename is provided, the program should output the computed hash.
Your program should handle errors in a meaningful manner.
Ensure the utilization of the appropriate OpenSSL algorithms.
Your task is to implement a file hashing and integrity verification program. You are not expected to build a graphical interface, but your program should use the command line interface. You will be asked to write a report documenting your implementation and how your program meets the specifications.</td></tr>
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Deliverables:	For your submission you are asked to provide the following:
	
	• **Code implementing file integrity checker (Weighting: 70%).**
	  o The code should include a Makefile and a full list of dependencies as part of a README file. Your README file should clearly say which parts of the assessment you have completed.
	  o The code should be formatted consistently, as per language norms, and commented appropriately.
	  o Grading Criteria:
	    ▪ Work with File I/O: 10 Marks
	    ▪ Apply Cryptographic Hashing: 26 Marks
	    ▪ Appropriate secure coding and OpenSSL Algorithms: 12 Marks
	    ▪ Appropriate error handling: 12 Marks
	    ▪ Good coding practice and code quality: 10 Marks
	• **A report (Weighting: 30%).**
	  o A report of no more than 1,500 words covering justification around algorithms, trade-offs, and awareness of security principles. Your report should be well-formatted, structured and titled appropriately.
	
	NB: If AI is used in any part of the coursework, you should clearly declare how it is used and in which parts.

Title of the Report	**Preserving File integrity**		
Student ID		**Level M**	**Weighting – 60 %**
Unit Director	Kopo Marvin Ramokapane	**Second marker**	Marvin Ramokapane

Report marker's name	**Implementation (70%)**	**Report (30%)**	**Total mark (weighted average)**
	Report mark (agreed mark of the two markers):		

Mark awarded (See University Generic Marking Criteria)	

Grade	**0-20 scale**	**0-100 scale**	**Criteria to be satisfied**
A	20 19 18	100 94 89	• Work would be worthy of dissemination under appropriate conditions • Mastery of advanced methods and techniques at a level beyond that explicitly taught • Ability to synthesise and employ in an original way ideas from across the subject • In group work, there is evidence of an outstanding individual contribution • Excellent presentation • Outstanding command of critical analysis and judgement

	17 16 15	83 78 72	• Excellent range and depth of attainment of intended learning outcomes • Mastery of a wide range of methods and techniques • Evidence of study and originality clearly beyond the bounds of what has been taught • In group work, there is evidence of an excellent individual contribution • Excellent presentation
B	14 13 12	68 65 62	• Attained all the intended learning outcomes • Able to use well a range of methods and techniques to come to conclusions • Evidence of study, comprehension and synthesis beyond the bounds of what has been explicitly taught • Very good presentation of material • Able to employ critical analysis and judgement • Where group work is involved there is evidence of a productive individual contribution
C	11 10 9	58 55 52	• Some limitations in attainment of learning objectives, but has managed to grasp most of them • Able to use most of the methods and techniques taught • Evidence of study and comprehension of what has been taught • Adequate presentation of material • Some grasp of issues and concepts underlying the techniques and material taught • Where group work is involved there is evidence of a positive individual contribution
D	8 7	48 45	• Limited attainment of intended learning outcomes • Able to use a proportion of the basic methods and techniques taught • Evidence of study and comprehension of what has been taught, but grasp insecure • Poorly presented • Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
	6	42	
	5	35	• Attainment of only a minority of the learning outcomes • Able to demonstrate a clear but limited use of some of the basic methods and techniques taught • Weak and incomplete grasp of what has been taught • Deficient understanding of the issues and concepts underlying the techniques and material taught
E	1-4	7-29	• Attainment of nearly all the intended learning outcomes deficient • Lack of ability to use at all or the right methods and techniques taught • Inadequately and incoherently presented • Wholly deficient grasp of what has been taught • Lack of understanding of the issues and concepts underlying the techniques and material taught
F	0	0	• No significant assessable material, absent or assessment missing a “must pass” component